

INFORMATION TECHNOLOGY INSTITUTE

LAB 4

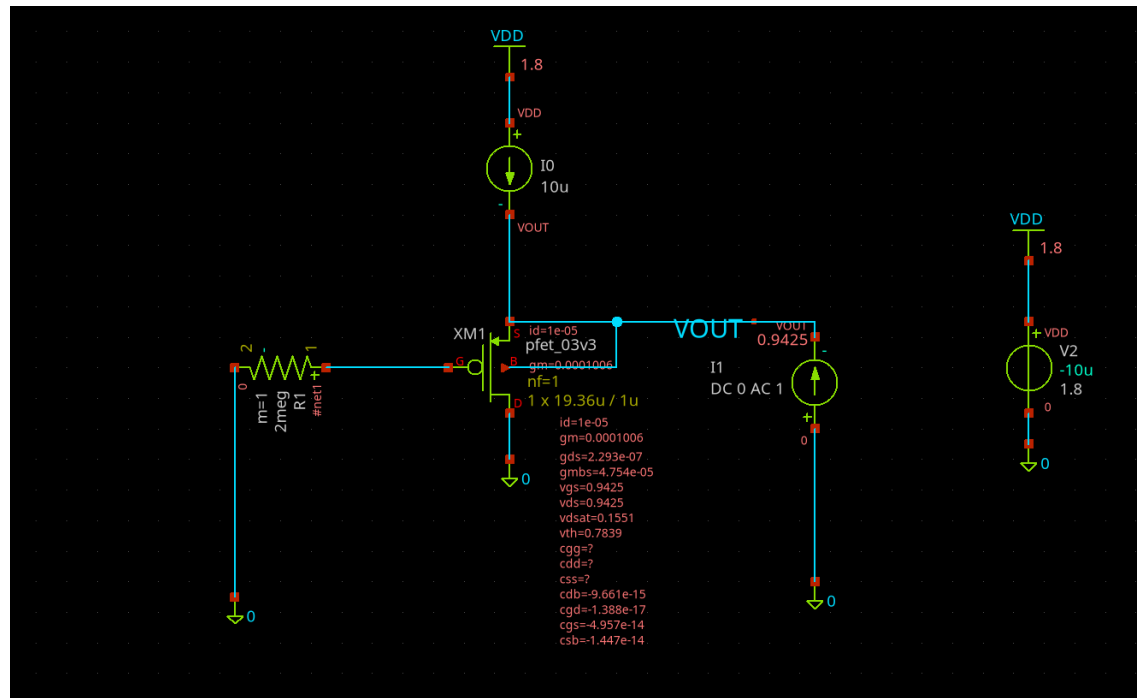
Common Drain Amplifier





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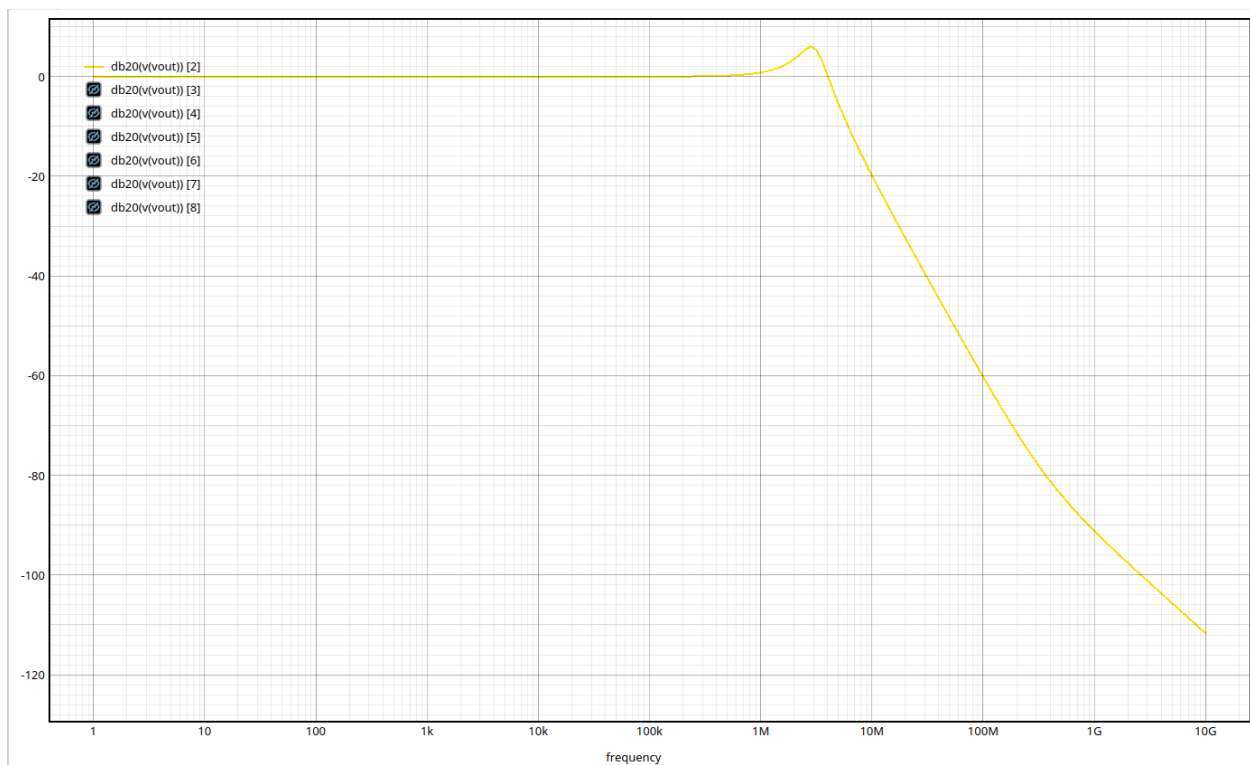
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Operates in saturation as the value of V_{ds} is greater than V_{dsat}

AC Analysis

Bode plot mag





It peaks around 6db

$CL = 2p$, Gain Peak (dB) = 5.9966

Analytically calculate the quality factor

$$Q = \frac{\sqrt{b_2}}{b_1} \approx \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{C_L}}$$

$g_m = 0.0001006$

$c_{gs} = 4.957e-14$

$c_{gd} = 1.388e-17$

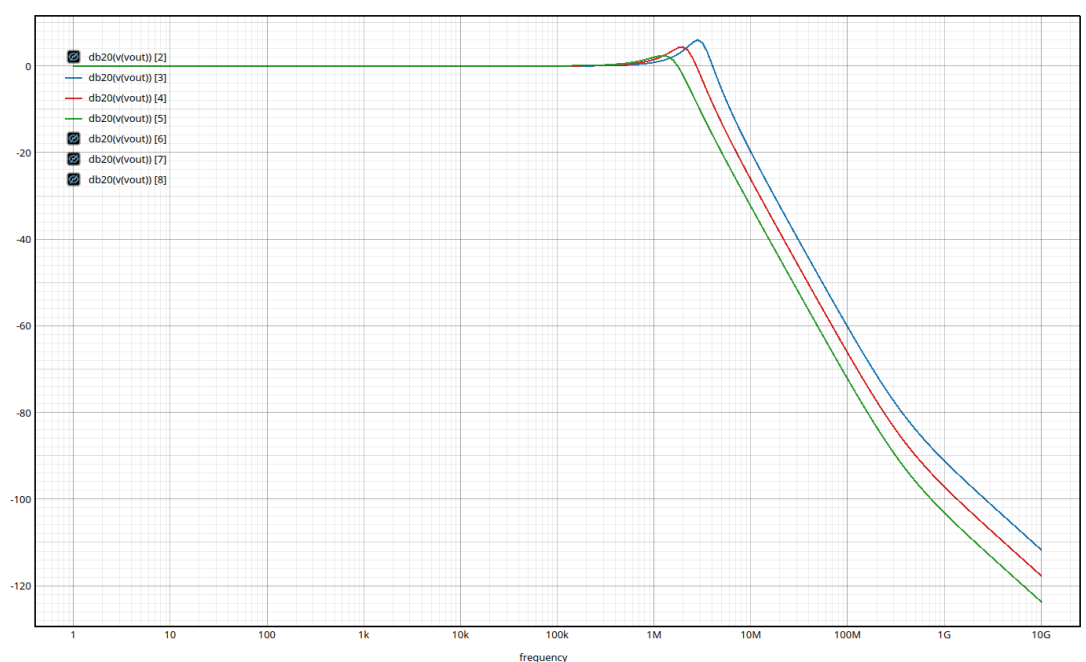
$c_L = 2e-12$

$R_{sig} = 2e6$

Substituting in this Rule we get $Q = 2.233$

which means the system is under damped as $Q > 0.5$ and this explains the peaking

parametric sweep: $CL = 2p, 4p, 8p$





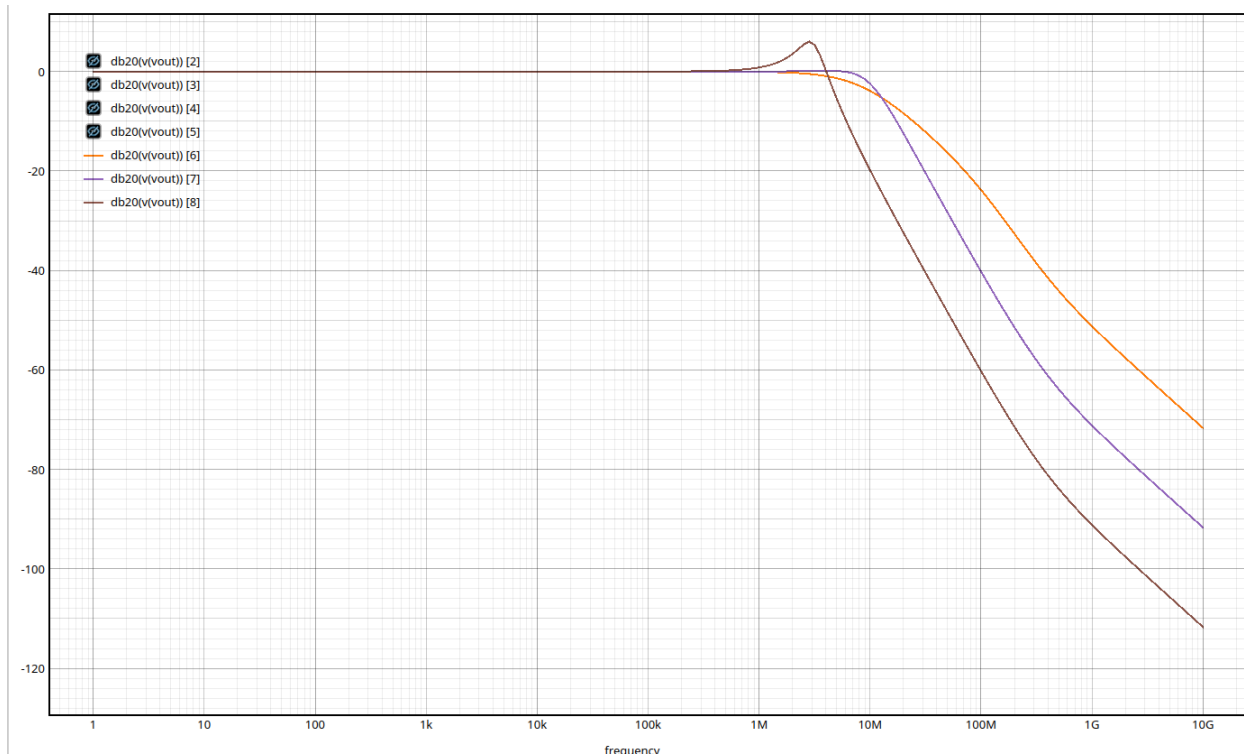
Peaking vs CL

CD Amplifier AC Sim Results

```
CL = 2p, Gain Peak (dB) = 5.9966  
CL = 4p, Gain Peak (dB) = 4.33344  
CL = 8p, Gain Peak (dB) = 2.3851
```

As CL increases the peaking decreases and this because the output pole decreases making the system slowly turning into critically damped system if we increased the CL even more.

Perform parametric sweep: R_{sig} = 20k, 200k, 2M.

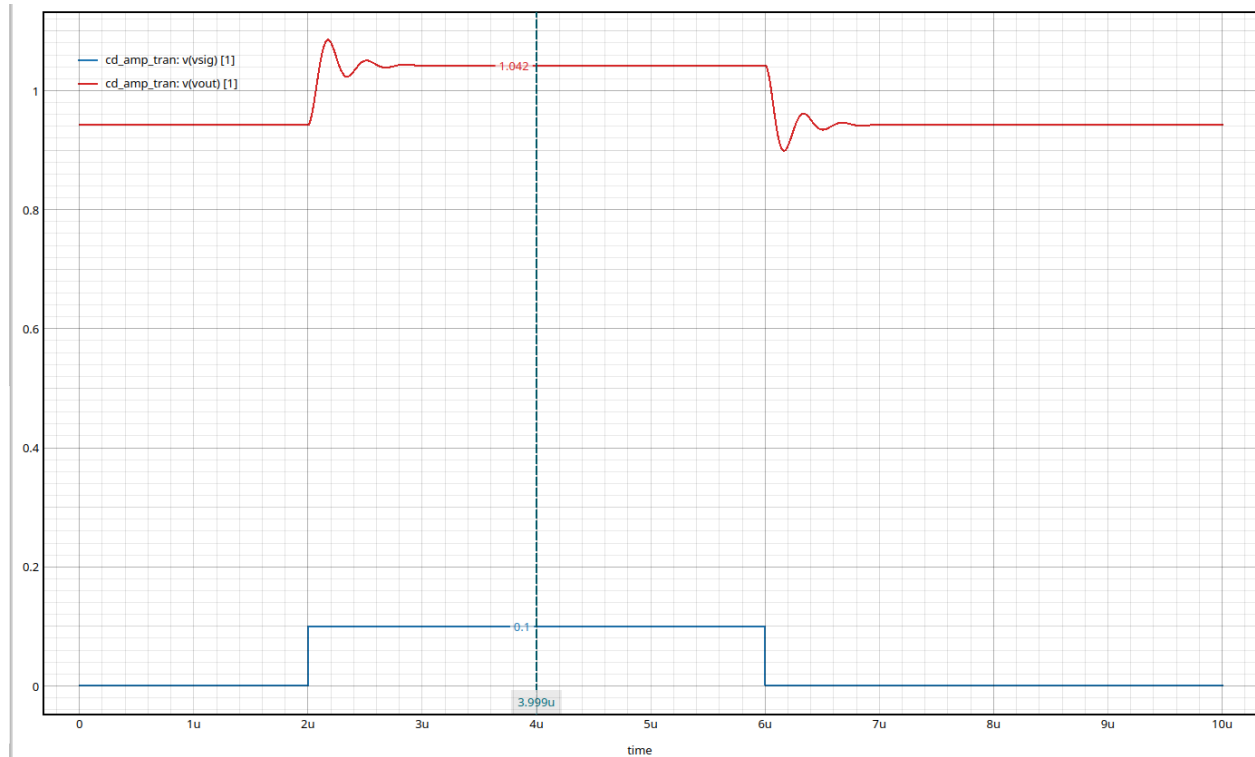


As R_{sig} increases the peaking increases and this because the input pole decreases, making the system slowly turning into underdamped system .

```
RES = 20k, Gain Peak (dB) = -0.0197724  
RES = 200k, Gain Peak (dB) = 0.121317  
RES = 2meg, Gain Peak (dB) = 5.9966
```



Transient Analysis



The DC shift is around 943mV which is equal to the $-V_{gs}$ of this mosfet

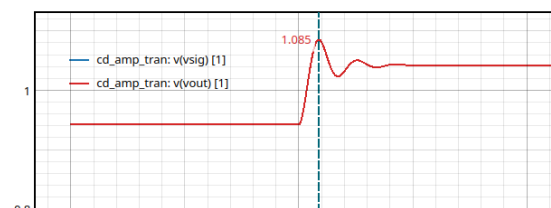
In order to shift it down we use a Nmos as the V_{gs} is positive

Ringing & Overshoot

peak is 1.085v

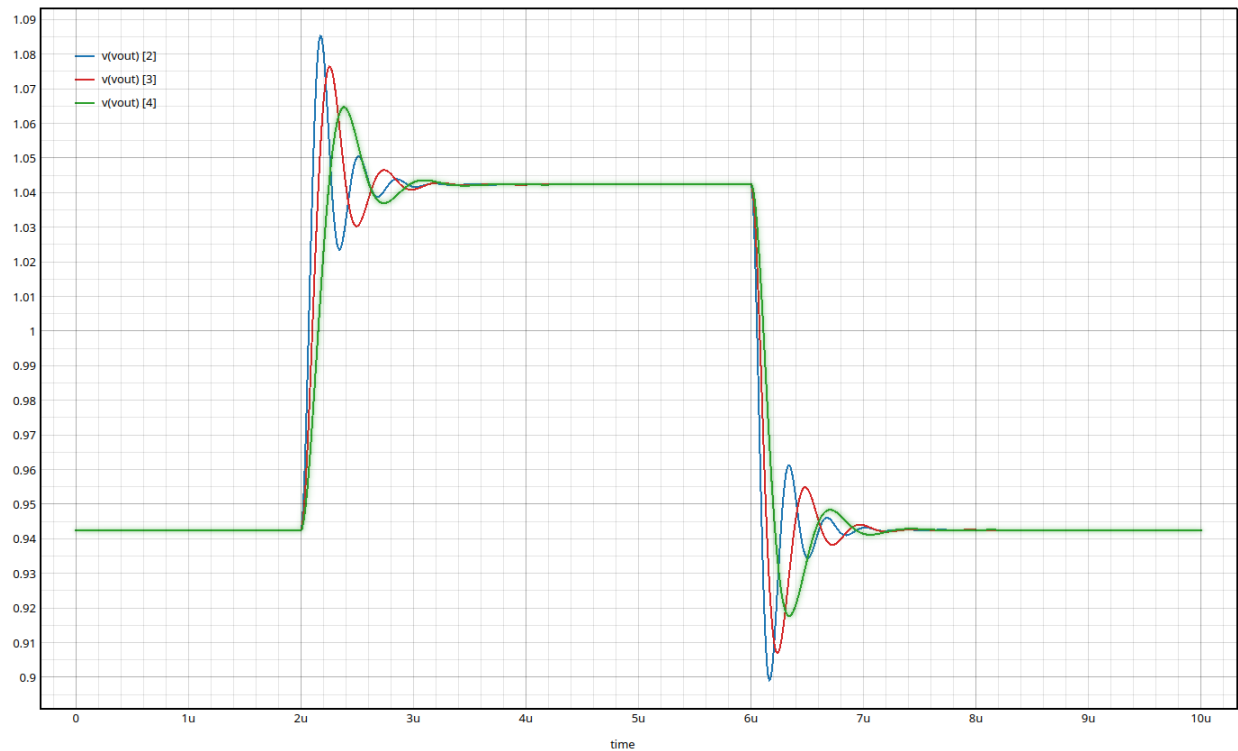
$$v_{\infty} = 1.042\text{v}$$

So overshoot is 43mV





parametric sweep: $CL = 2p, 4p, 8p$.

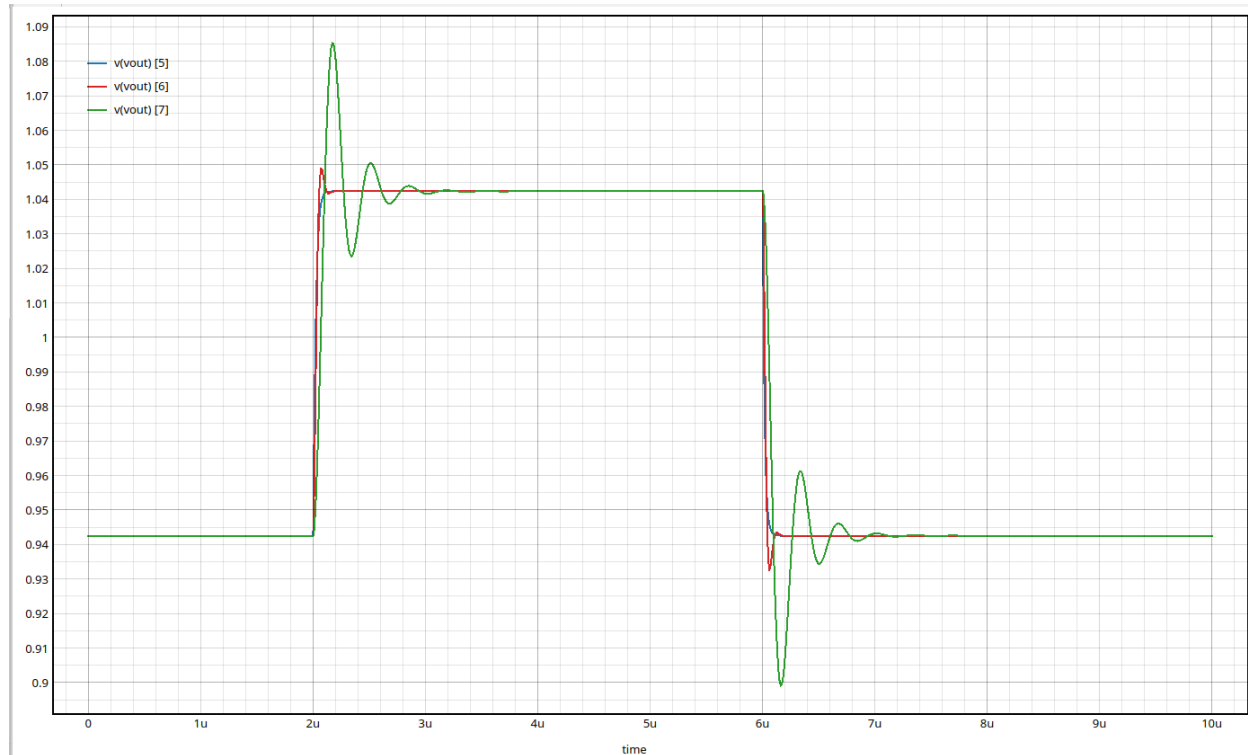


```
CD Amplifier Transient Sim Results  
CL = 2p, Overshoot (%) = 43.0232  
CL = 4p, Overshoot (%) = 34.1077  
CL = 8p, Overshoot (%) = 22.4211
```

As CL increases the over shoot decreases as result of the system becoming less underdamped and transitioning to critically damped and as the Q decreases and also as CL controls the output pole.



parametric sweep: $R_{sig} = 20k, 200k, 2M$.



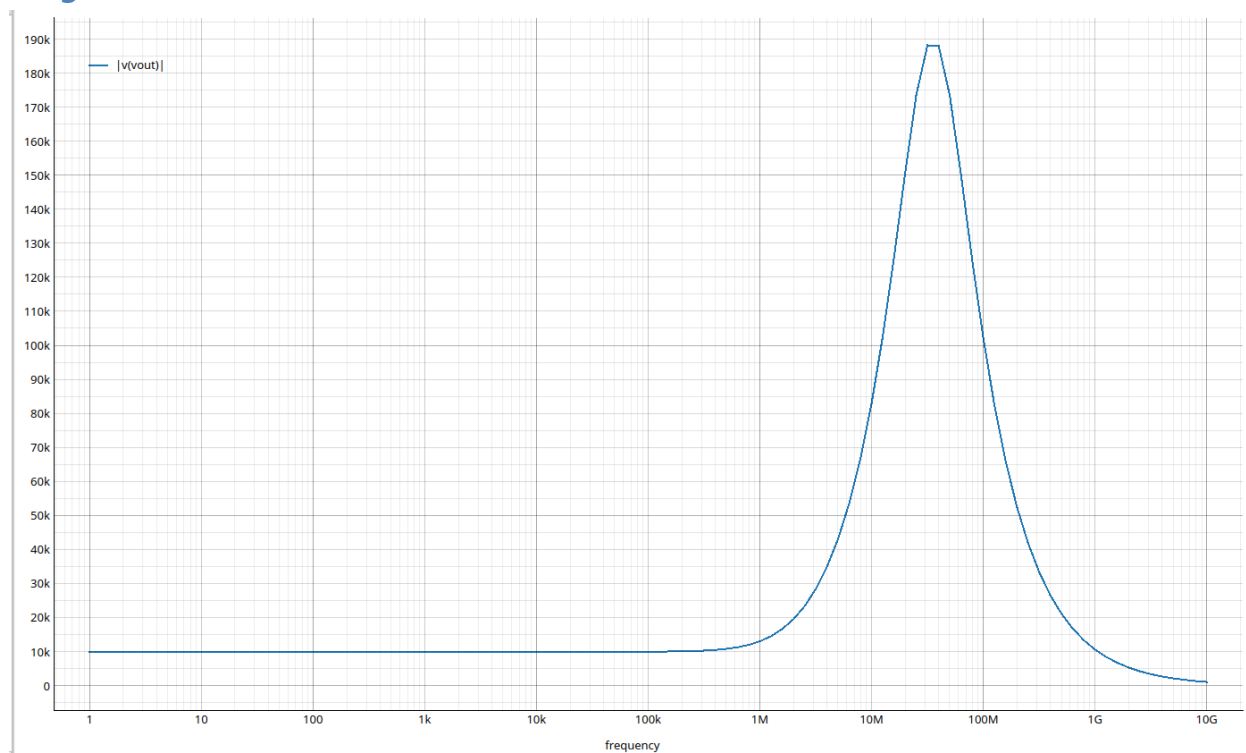
```
Rsig = 20k, Overshoot (%) = 0  
Rsig = 200k, Overshoot (%) = 6.58342  
Rsig = 2meg, Overshoot (%) = 43.0232
```

As the R_{sig} increases the quality factor increases and the system start transitioning to under damped system in which the overshoot increases

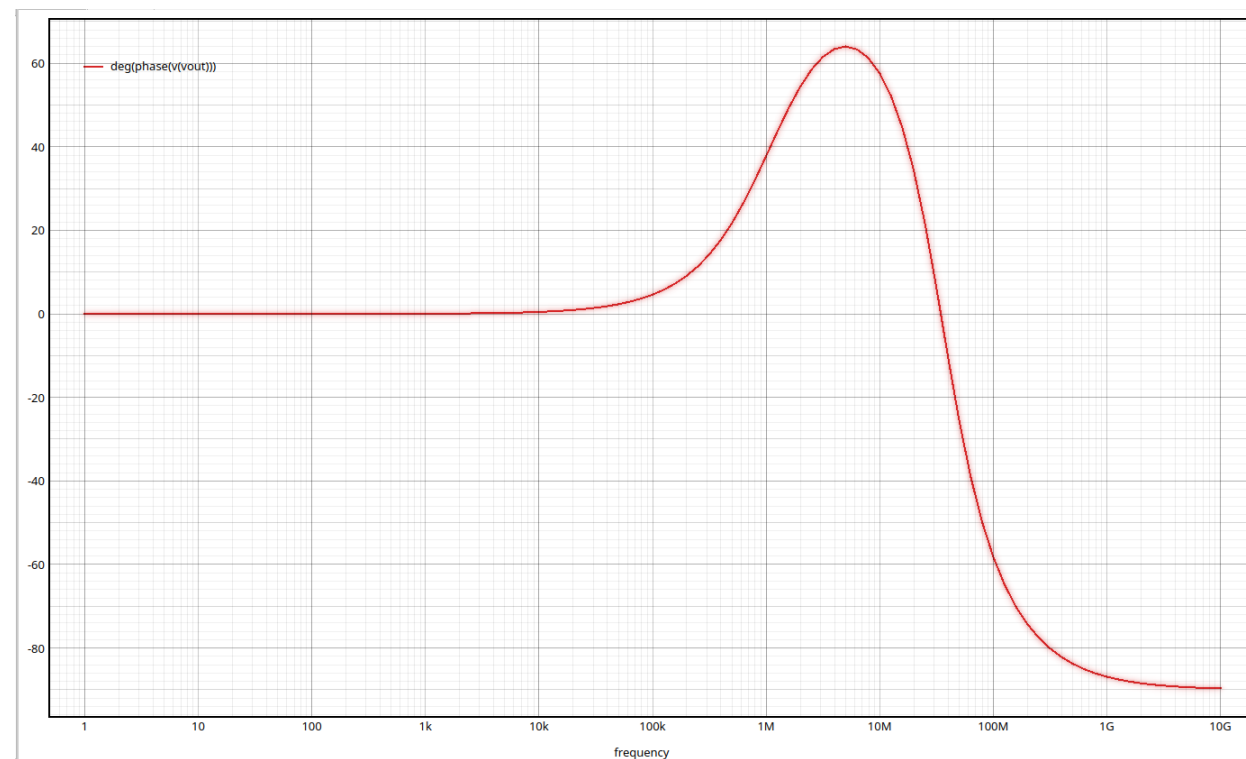


Z_{out} (Inductive Rise)

Mag



phase





Inductive rise

It is noticeable and it is because at low frequencies $z_{out}=1/g_m$ but when ω increases z_{out} approaches R_{sig} as at high frequencies C_{gs} makes the output node see the resistance of R_{sig}

$$Z_{out} = \frac{1}{g_m} \left(\frac{1 + sR_{sig}C_{gs}}{1 + s\frac{C_{gs}}{g_m}} \right)$$

Why does it fall

Because at frequencies the C_{gd} decreases and shunts the R_{sig}

Analytical calculations of poles and zeros

Using an unapproximated approach

$$Z_{out} = \frac{1}{g_m} \frac{1 + sR_{sig}(C_{gd} + C_{gs})}{\left(1 + s\frac{C_{gs}}{g_m}\right)(1 + sR_{sig}C_{gd})}$$

$$g_m = 0.0001006$$

$$R_{sig} = 2e6$$

$$C_{gs} = 4.957e-14$$

Zero

$$\omega_z = \frac{1}{R_{sig}(C_{gs} + C_{gd})}$$

$$F_z = 1.605 \text{ MHz}$$

Pole

$$\omega_p = \frac{g_m}{C_{gs}} = \frac{1.006 \times 10^{-4}}{4.957 \times 10^{-14}} \approx 2.029 \times 10^9 \text{ rad/s}$$

$$F_p = 323 \text{ MHz}$$

At low frequency it is $1/g_m = 9940 \Omega$



At high frequency it is approximately $=R_{sig}=2\text{meg}\Omega$

Sim results

```
CD Amplifier Zout Sim Results  
Zout @ 1kHz = 9917.68  
Zout peak (max) = 188181
```

	Hand analysis	Simulation extracted from bode plot
Zout at High frequencies	2Meg Ω	188.2K Ω
Zout at Low frequencies	9940 Ω	9918 Ω
F pole	323 MHz	32 MHz
F zero	1.605MHz	1.3MHz

